PHY 412: Statistical Physics

26-26 Semester II

Assignment – 1

Problems in black are for submission; Blue ones are for practice. Due date: January 28th in Class

1. The three continuous probability distributions most seen in physics are: (i) *uniform:* $\rho(x) = 1$ for $0 \leq x < 1$, $\rho(x) = 0$ otherwise, (ii) *Exponential:* $\rho(x) = \frac{e^{-x/\tau}}{\tau}$ for $x \geq 0$, and (iii) *Gaussian:* $\rho(x) = \frac{e^{-x^2/2\sigma^2}}{\sqrt{2\pi}\sigma}$
   (a) What is the probability that a random number uniform on [0, 1) will happen to lie between x = 0.7 and x = 0.75?
   (b) That the waiting time for a radioactive decay of a nucleus will be more than twice the exponential decay time τ?
   (c) That your score on an exam with a Gaussian distribution of scores will be greater than 2σ above the mean?
   (d) Show that these probability distributions are normalized: $\int \rho(x)dx = 1$.
   (e) What is the mean x_0, and standard deviation $\sqrt{\int (x - x_0)^2 \rho(x)dx}$, of each distribution?
   (f) Draw a graph of the probability distribution of the sum $x + y$ of two random variables drawn from a uniform distribution on [0, 1). Argue in general that the sum $z = x + y$ of random variables with distributions $\rho_1(x)$ and $\rho_2(y)$ will have a distribution given by $\rho(z) = \int \rho_1(x)\rho_2(z - x)dx$ (the convolution of ρ with itself).

2. A two-dimensional $L \times L$ box with hard walls contains a gas of N distinguishable hard disks of radius $r \ll L$. It's a dilute system, *i.e.*, the summed area $N\pi r^2 \ll L^2$. Let A be the effective area allowed for the disks in the box, $A = (L - 2r)^2$. Draw the excluded volume around a disk and around the wall.
   (a) What is the allowed $2N$-dimensional volume in configuration space ($\Omega(q_i)$) of allowed configurations of hard disks, in this dilute limit?
   (b) Show that at $r = 0$, you reproduce 2D ideal gas result $\Omega(q_i) = L^{2N}$.

3. Consider the velocity of a gas particle in one dimension ($-\infty < v < \infty$).
   (a) Find the unbiased probability density $\rho_1(v)$, subject only to the constraint that the average speed is c, that is, $\langle |v| \rangle = c$.
   (b) Now find the probability density $\rho_2(v)$, given only the constraint of average kinetic energy, $\langle mv^2/2 \rangle = mc^2/2$.
   (c) Which of the above statements provides more information on the velocity? Quantify the difference in information in terms of $I_2 - I_1 \equiv (\langle \ln \rho_2 \rangle - \langle \ln \rho_1 \rangle)/\ln 2$.

4. Show that the phase space volume remains invariant under coordinate transformation.
   (a) Verify explicitly the invariance of the volume element $d\omega$ of the phase space of a single particle under transformation from the Cartesian coordinates (x, y, z, p_x, p_y, p_z) to the spherical polar coordinates $(r, \theta, \phi, p_r, p_\theta, p_\phi)$.
   (b) The foregoing result seems to contradict the intuitive notion of "equal weights for equal solid angles," because the factor $\sin\theta$ is invisible in the expression for $d\omega$. Show that if we average out any physical quantity, whose dependence on p_θ and p_ϕ comes only through the kinetic energy of the particle, then as a result of integration over these variables we do indeed recover the factor $\sin\theta$ to appear with the sub-element $(d\theta\, d\phi)$.

5. Further, show that the phase space volume remains invariant under a canonical transformation of the generalized coordinates from one set to another. The generalized coordinates of a simple pendulum are the angular displacement θ and the angular momentum $ml^2\dot{\theta}$. Show, both mathematically and graphically, the nature of the corresponding trajectories in the phase space of the system and prove that

the area A enclosed by a trajectory is equal to the product of the total energy E and the time-period τ of the pendulum.

6. Characteristic functions: calculate the characteristic function, the mean, and the variance of the following probability density functions:
   (a) Laplace $= \frac{1}{2a} \exp\left(-\frac{|x|}{a}\right)$.
   (b) Cauchy $\rho(x) = \frac{a}{\pi(x^2+a^2)}$.
   The following two probability density functions are defined for $x \geq 0$. Compute only the mean and variance for each.
   (c) Rayleigh $\rho(x) = \frac{x}{a^2} \exp\left(-\frac{x^2}{2a^2}\right)$.
   (d) Maxwell $\rho(x) = \sqrt{\frac{2}{\pi}\frac{x^2}{a^3}} \exp\left(-\frac{x^2}{2a^2}\right)$.

7. Consider any probability density $\rho(x)$ for $(-\infty < x < \infty)$, with mean λ, and variance σ^2. Show that the total probability of outcomes that are more than $n\sigma$ away from λ is less than $1/n^2$, that is,
$$\int_{|x-\lambda| \geq n\sigma} dx\, \rho(x) \leq \frac{1}{n^2}$$
   Hint: Start with the integral defining σ^2, break it up into parts corresponding to $|x - \lambda| \geq n\sigma$, and $|x - \lambda| < n\sigma$.

8. The current I across a diode is related to the applied voltage V via $I = I_0\,[\exp(eV/k_B T) - 1]$. The diode is subject to a random potential V of zero mean and variance σ^2, which is Gaussian distributed. Find the probability density $\rho(I)$ for the current I flowing through the diode. Find the most probable value for I, the mean value of I, and indicate them on a sketch of $\rho(I)$.

9. A mirror is plated by evaporating a gold electrode in vacuum by passing an electric current. The gold atoms fly off in all directions, and a portion of them sticks to the glass (or to other gold atoms already on the glass plate). Assume that each column of deposited atoms is independent of neighboring columns, and that the average deposition rate is d layers per second.
   (a) What is the probability of m atoms deposited at a site after a time t? What fraction of the glass is not covered by any gold atoms?
   (b) What is the variance in the thickness?

10. Alternative derivation of grand canonical distribution: Consider M copies of a system distributed among discrete energy states such that m_i copies are in ϵ_i state. Obviously, $\sum_i m_i = M$. Let N_i be the number of particles in each of m_i systems. Find the most probable distribution for $\{m_i\}$ given that the average energy and the average particle number are fixed.

11. Suppose that a set of N non-interacting free particles, each with charge q and mass m, are moving in a uniform magnetic field $\boldsymbol{B}$ inside a box of size V. If $\boldsymbol{A}$ is the vector potential of the magnetic field, then the Hamiltonian of the system is $H = \frac{1}{2m}\sum_{i=1}^{N}(\boldsymbol{p}_i - q\boldsymbol{A}(\boldsymbol{x}_i)/c)^2$. Show that the average magnetization of the system vanishes.

12. In constant pressure ensemble, E, V, and S are random variables. Show the following:
   (a) $\sigma_V^2 = k_B T \kappa_T \langle V \rangle$, where κ_T is the isothermal compressibility.
   (b) $\sigma_E^2 = k_B T^2 (\partial\langle E\rangle/\partial T)_P + k_B T P (\partial\langle E\rangle/\partial P)_T$.
   (c) $\sigma_S^2 = k_B C_P$. [You may need to make use of the first law of thermodynamics.]